Exercise 07: Relational database design and normalisation

Task 1*: Normalisation

Given the following relations R and S in the first normal form with functional dependencies $F = F_C$:

$$\begin{aligned} R &= (A,B,C,D,E,F) \; with \\ A,B &\rightarrow C,D,E \\ D &\rightarrow F \; and \\ &\{A,B\} \text{ is Key.} \end{aligned} \qquad \begin{aligned} S &= (V,W,X,Y,Z) \; with \\ V &\rightarrow W,Z \\ W,Z &\rightarrow V,X,Y \\ Y &\rightarrow Z \; and \\ &\{V\},\{W,Z\},\{W,Y\} \text{ are keys.} \end{aligned}$$

1. *Show as briefly as possible that R does not correspond with the third normal form (3NF).
2. *Through decomposition, split R into relations that correspond with the Boyce–Codd normal form (BCNF). Are the resulting relations dependency preserving and lossless join decomposition?
3. Split R using the synthesis process. Are the resulting relations dependency preserving and a lossless join decomposition?
4. *Show as briefly as possible that S does not correspond with the Boyce-Codd normal form (BCNF).
5. *Through decomposition, split S into relations that correspond with the Boyce-Codd normal form (BCNF). Are the resulting relations dependency preserving and lossless join decomposition? Is the decomposition useful?
6. Split S using the synthesis process. Are the resulting relations dependency preserving and lossless join decomposition?

Task 2: Synthesis process

Given the following relation R and functional dependencies F:

$$\begin{aligned} R &= (A,B,C,D,E,F) \\ A &\rightarrow B,C \\ D &\rightarrow E,F \end{aligned}$$

1. Obviously R does not correspond with the second normal form. The only key is $\{A,D\}$ and all prime attributes only depend on a real subset of the key. At which point in the synthesis process will a difficulty arise with the runtime?
2. If you used the simplified synthesis procedure with the additional rule $A,B,C,D,E,F \rightarrow \delta$, what is the improvement and what result is to be expected?
3. Carry out the synthesis process for R using the cover-algorithm to generate a superkey.

Task 3: Comparison of synthesis and decomposition

Given the following relations S_1 and S_2 in the first normal form with functional dependencies:

$$\begin{aligned} S_1 &= (A, P, H, R, O, D, I, T, E)\ with \\ R &\rightarrow O, D \\ O &\rightarrow A, H, P, R\ and \\ &\{I, T, E, R\},\ \{I, T, E, O\} \text{ are keys.} \end{aligned}$$

$$\begin{aligned} S_2 &= (A, P, H, R, O, D, I, T, E)\ with \\ R &\rightarrow O \\ O &\rightarrow A, P, H, R, D\ and \\ &\{I, T, E, R\},\ \{I, T, E, O\} \text{ are keys.} \end{aligned}$$

1. Through decomposition, split S_1 and S_2 into relations in the Boyce-Codd normal form (BCNF). Can this be done?
2. Decompose the relational schema using the synthesis process.
3. Carry out the synthesis process by using the cover-algorithm to find a superkey.
4. So now, which process is better: decomposition, synthesis or the simplified synthesis process?